Operations

Week 5 – Live Class

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THREE MEASURES OF PROCESS PERFORMANCE

Cycle Time (time/unit)

- Average time between the output of successive units (e.g. 5 min/unit)
- **Flow rate** or **throughput rate**) is the inverse of the cycle time (e.g. 1/5 units /min = 0.2 units/min)

Flow time or manufacturing lead time or service lead time

- The average amount of time it takes a unit to get through the process (includes waiting time).

Inventory or Work in process (WIP)

- Average number of units in the process at any point in time (includes the units waiting to be processed).

Variability sources and its impact

Sources of variability

- Demand variability
- Variability in processing times

Impact of variability

- Utilisation/efficiency (Original-i-tee example)
- Higher Inventory levels (Every Season example)
- Waiting times (Topic 2)
- Throughput losses (Topic 3)

Why do queues build?

Two primary reasons

1. **Implied utilisation above 100%**
   - Demand rate exceeds supply rate for specific periods of time.
   - Examples: Queues to enter a music concert, queues to enter an airplane.

2. **Variability**
   - Waiting times can exist even if there is enough capacity on average (i.e. implied utilisation is below 100%)
   - Examples: queues at a grocery store, client service support, restaurants, emergency services, web server congestion, etc.

An experiment: An unrealistic call center

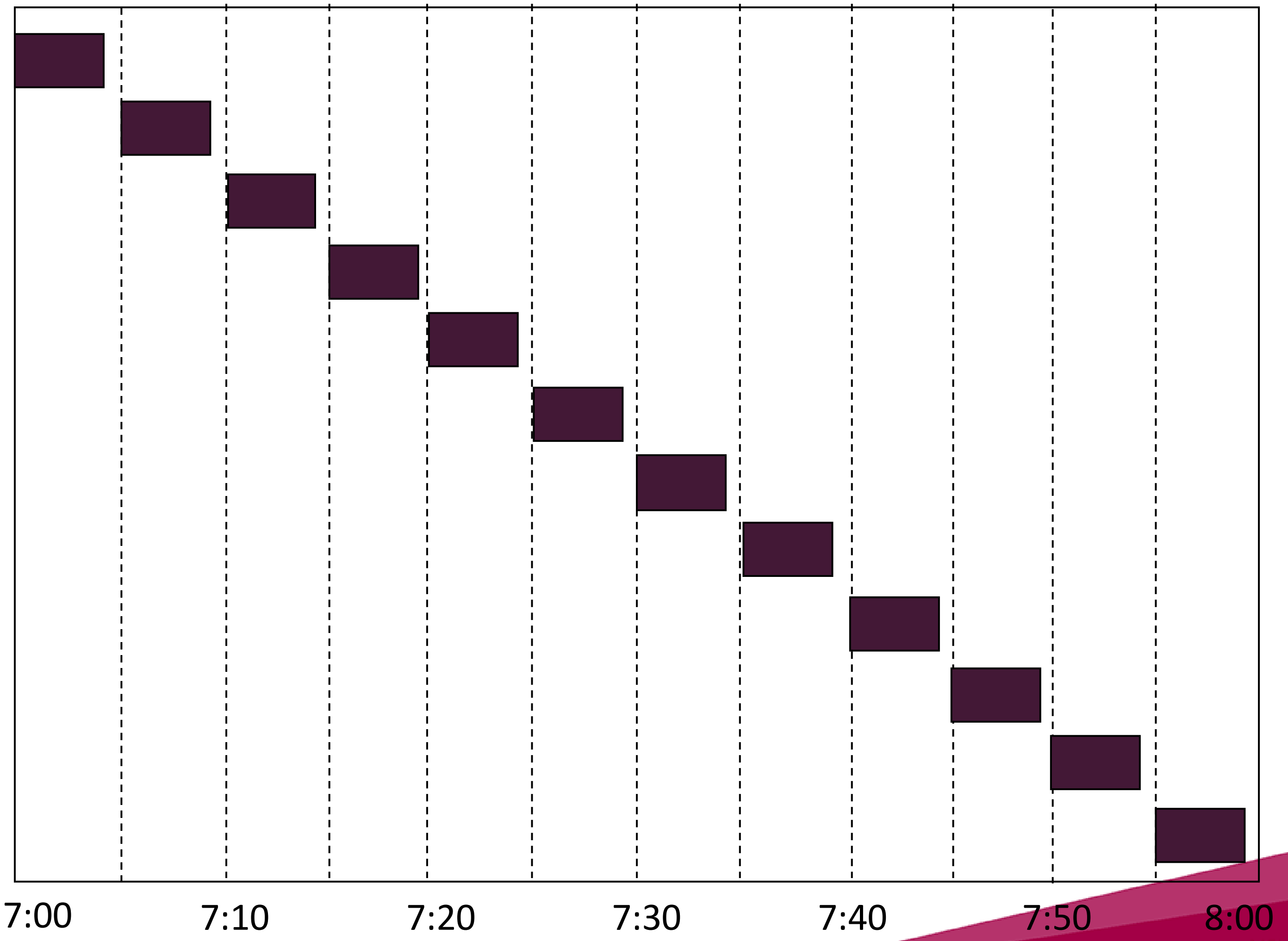
One employee works from 7am to 8am
Call duration: 4 minutes
Number of calls per hour: 12 (i.e. one call every 5 minutes)

Capacity : 15 calls/hour
Demand: 12 calls/hour
Utilisation = 80%

FACT:
If the calls arrive exactly every 5 minutes then NO client needs to wait.

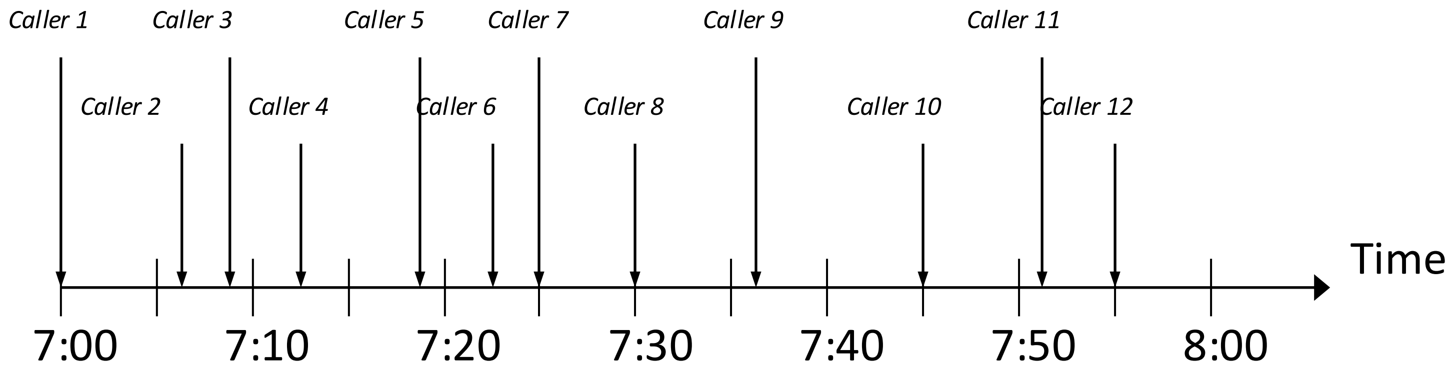
A thought experiment: An unrealistic call center

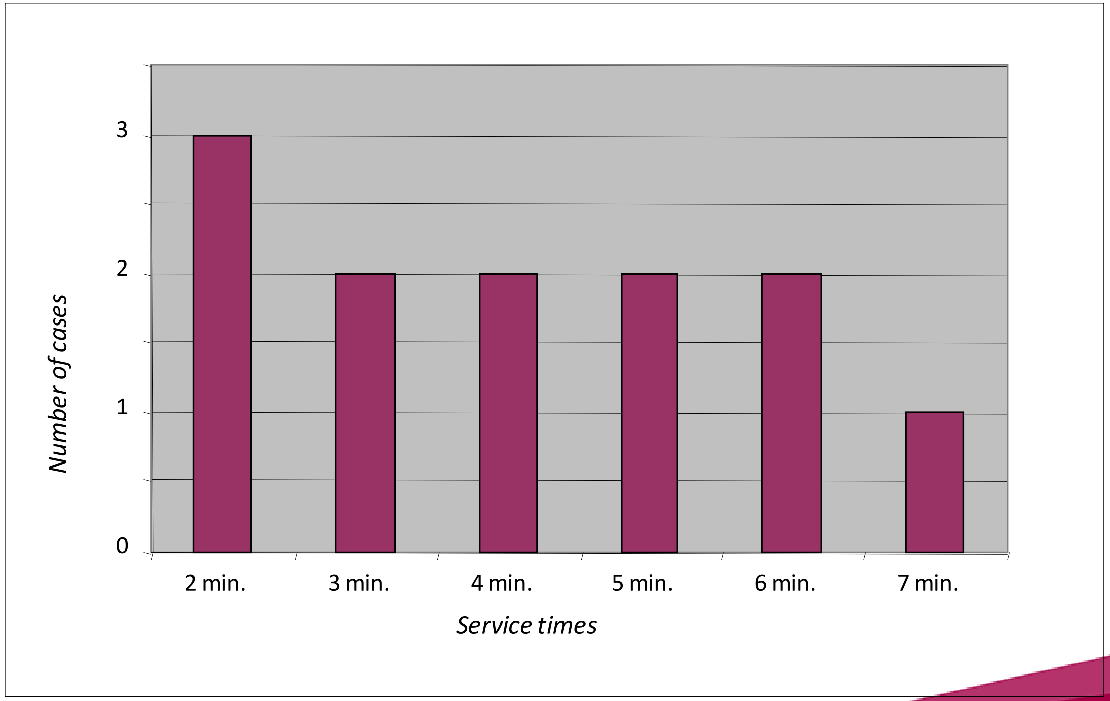
Caller	Arrival Time	Service Time
1	0	4
2	5	4
3	10	4
4	15	4
5	20	4
6	25	4
7	30	4
8	35	4
9	40	4
10	45	4
11	50	4
12	55	4



A More Realistic Service Process

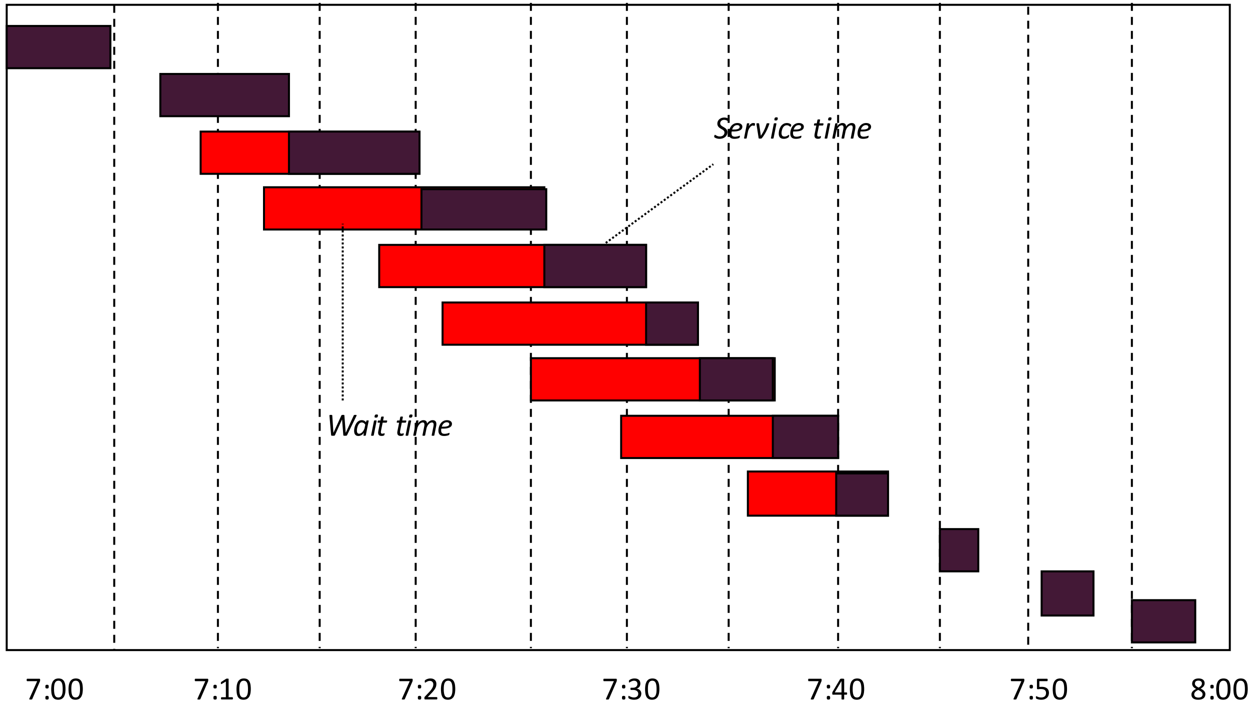
Caller	Arrival Time	Processing Time
1	0	5
2	7	6
3	9	7
4	12	6
5	18	5
6	22	2
7	25	4
8	30	3
9	36	4
10	45	2
11	51	2
12	55	3

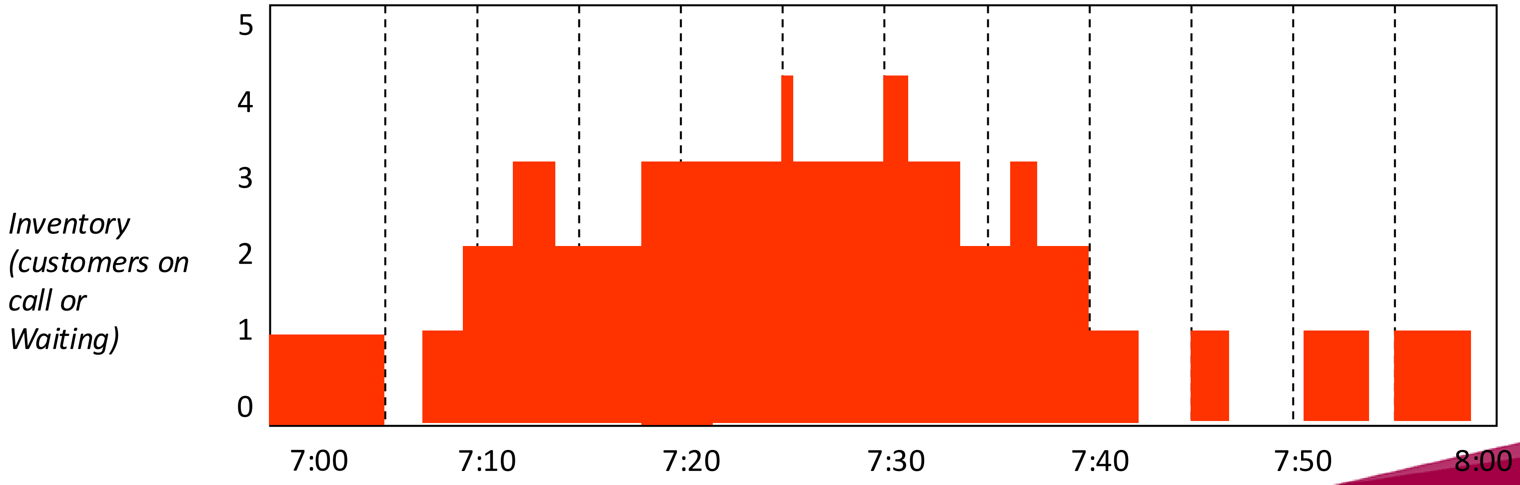




Variability Leads to Waiting Time

Caller	Arrival Time	Service Time
1	0	5
2	7	6
3	9	7
4	12	6
5	18	5
6	22	2
7	25	4
8	30	3
9	36	4
10	45	2
11	51	2
12	55	3





Simulation Results (Tea machine)

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	Avg arrivals per hour	Utilisation rate	Avg waiting time
Scenario 1	2 / hr	16.66%	0.5 min
Scenario 2	5/hr	41.66%	1.75 min
Scenario 3	10/hr	83.33%	12.71 min
Scenario 4	11/hr	91.66%	28.15 min

Conclusions about the simulation experiment

- Waiting times appear even if the utilisation of the tea machine is less than 100%.
- The waiting time is NOT directly proportional to the average number of employees arrivals per hour.
- Thus, the waiting time is NOT directly proportional to the utilisation rate.
- Waiting times tend to blow up when the utilisation rate approaches 100%

Quantifying variability in processes

Demand

a: Average interarrival time

σ_a: Standard deviation of the interarrival time

CV_a = σ_a / a

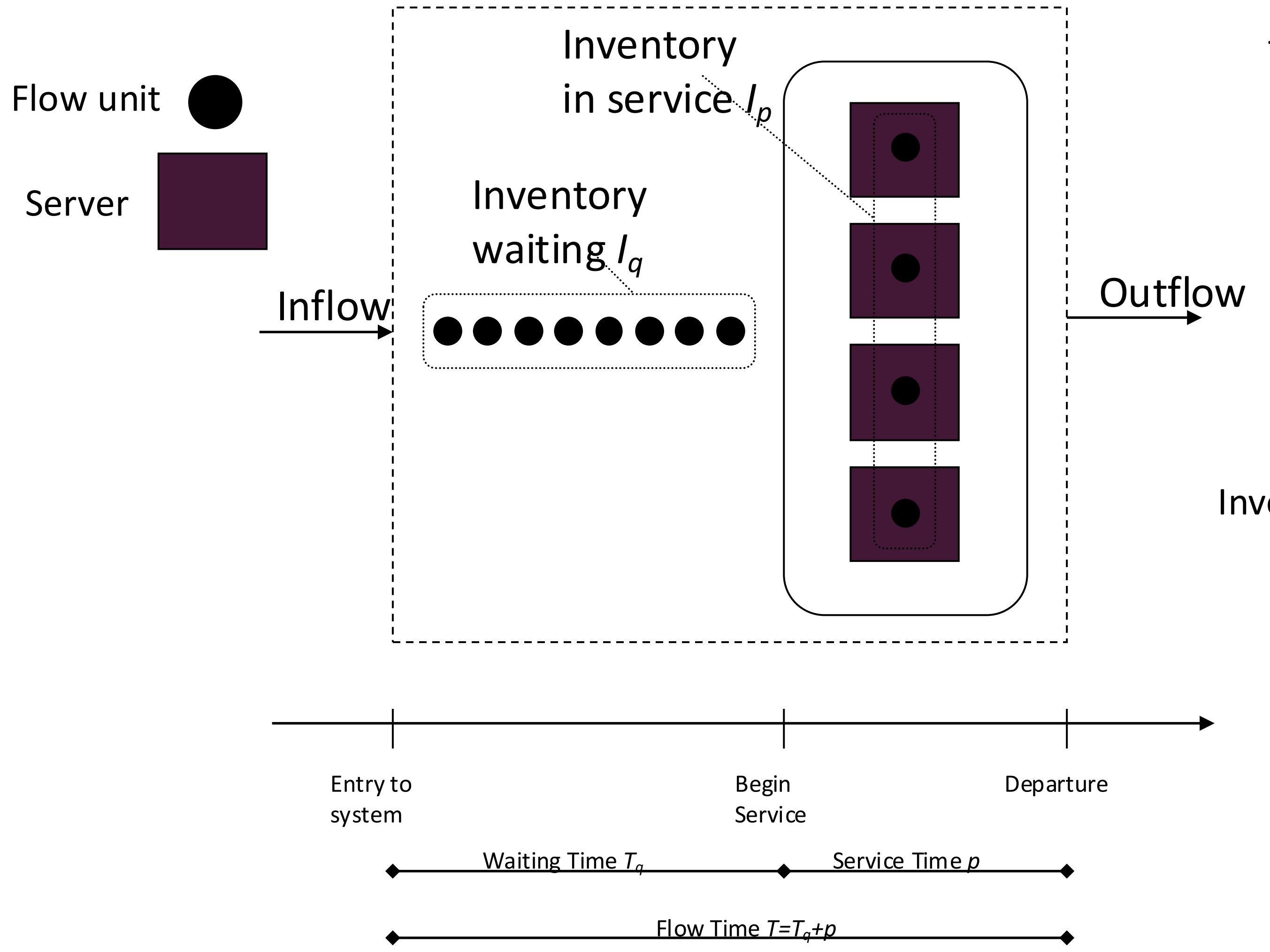
Processing step

p: Average processing time

σ_p: Standard deviation of the processing time

CV_p = σ_p / p

Summary of Queuing Analysis



$$u = \frac{1/a}{m * 1/p} = \frac{p}{am}$$

Time related measures

$$T_q \approx \left(\frac{p}{m}\right) * \left(\frac{u^{\sqrt{2(m+1)}-1}}{1-u}\right) * \left(\frac{CV_a^2 + CV_p^2}{2}\right)$$

$$T = T_q + p$$

Inventory related measures (Flow rate=*1/a*)

$$I_q = \frac{1}{a} * T_q$$

$$I_p = u * m$$

$$I = I_p + I_q$$

Waiting times: Excel Calculator

Number of servers	1
Average interarrival time	15.00
Average processing time	12.00
Coeff. of Variation for arrival times	1.00
Coeff. of Variation for processing time	0.00

Average server utilisation	**80.00%**
Waiting time	**24.00**

$$\text{Inventory in the queue} = I_q = \frac{T_q}{a}$$

$$\text{Inventory in the processing step} = I_p = \frac{p}{a}$$

Security on campus

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The North East University walking escort service is operated 24 hours a day, seven days a week. Students request a walking escort by phone. Requests for escorts are received, on average, every five minutes with a coefficient of variation of one.

After receiving a request, the dispatcher contacts an available escort (via a mobile phone), who immediately proceeds to pick up the student and walk them to their destination. If there are no escorts available (that is, they are all either walking a student to their destination or walking to pick up a student), the dispatcher puts the request in a queue until an escort becomes available.

Not including the time that a student waits for the escort to arrive after calling to place a request, the trip takes, on average, 25 minutes: From the time the escort meets the student to walk the student to the desired location (the coefficient of variation of this time is also one). Currently, the university has eight security officers who work as walking escorts.

Q8.8

(a) How many security officers are, on average, available to satisfy a new request?

(b) How much time does it take on average from the moment a student calls for an escort to the moment the student arrives at her/his destination?

Q8.8 solution

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(a) m = 8

p = 25 mins

a = 5 mins

Utilisation = $\frac{p}{a \times m} = \frac{25}{5 \times 8} = 62.5\%$

so idle officers are 100 - 62.5 = 37.5% of 8 officers = 3 officers

(b) $CV_a = 1$

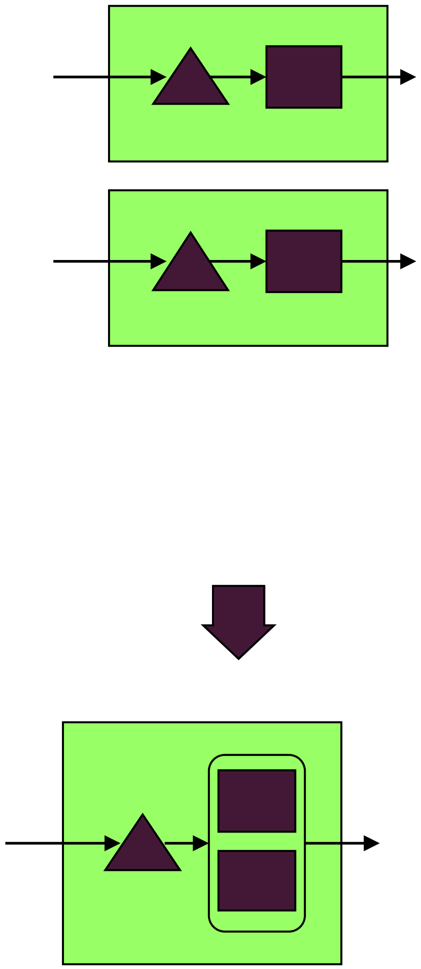
$CV_p = 1$

$$T_q = \left(\frac{Processing\ Time}{m}\right) \times \left(\frac{Utilization^{\sqrt{2(m+1)}-1}}{1-Utilization}\right) \times \left(\frac{{CV_a}^2 + {CV_p}^2}{2}\right)$$

$$T_q = \left(\frac{25}{8}\right) \times \left(\frac{0.625^{\sqrt{2(8+1)}-1}}{1-0.625}\right) \times \left(\frac{1+1}{2}\right) = 1.815\ mins$$

Once the officer arrives, it still takes 25 minutes to take them to their desired location, so Total time = 25 + 1.8 = 26.8 mins

Discussion: Merging processes



- Suppose we have two processes which are identical. Each one has one server.
- The processing time is p= 1 min
- Each one has a demand of 30 customers per hour.
- Cv_a=1, Cv_p =1

What happens if we merge them?

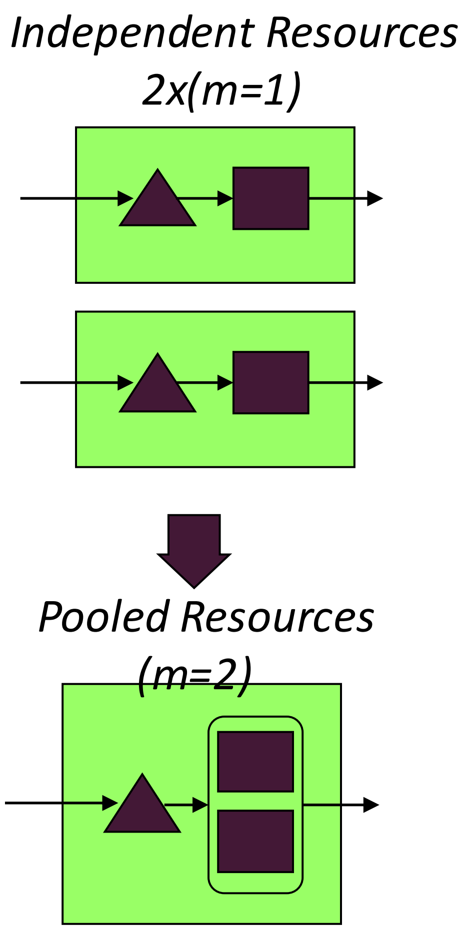
<u>Compare waiting times using the waiting time calculator</u>

Increases

Same

Decreases

Impact of Pooling: Applications



Applications

- Call centres
- Service support
- Cashiers payments
- Clinics

Advantages

Economies of scale (easy to see benefit)

Reduction in waiting times

Disadvantages

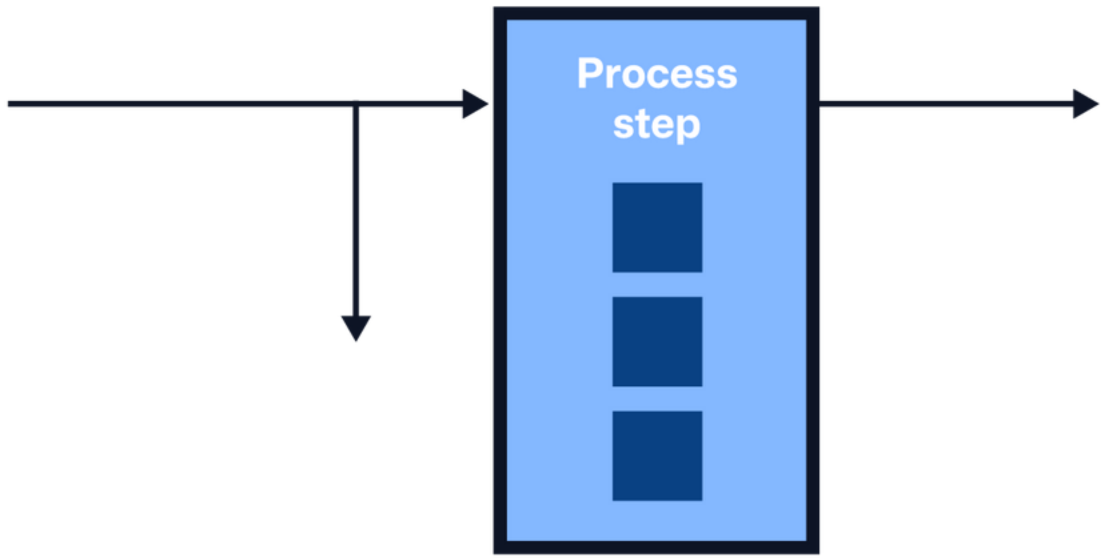
many points of contact

Throughput losses

Sometimes customers are **not willing to wait** and instead leave without receiving service.

Examples:

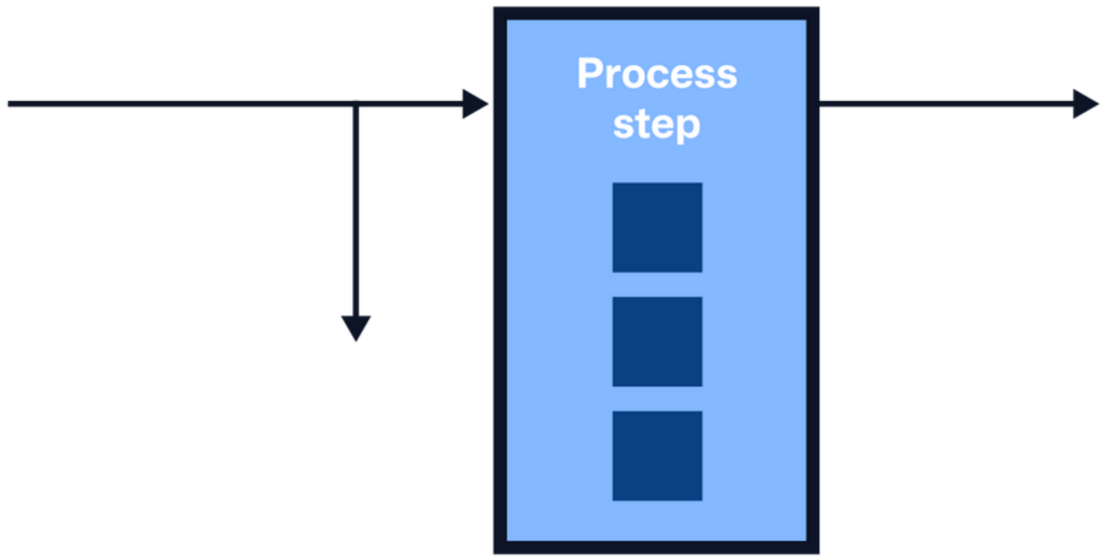
- Restaurants
- Physical retail stores
- Online retail stores



Throughput losses

Relevant information:

- Average inter-arrival time: *a*
- Average processing time: *p*
- Number of servers: *m*



Erlang loss probability

The probability that all *m* servers are busy (P_m).

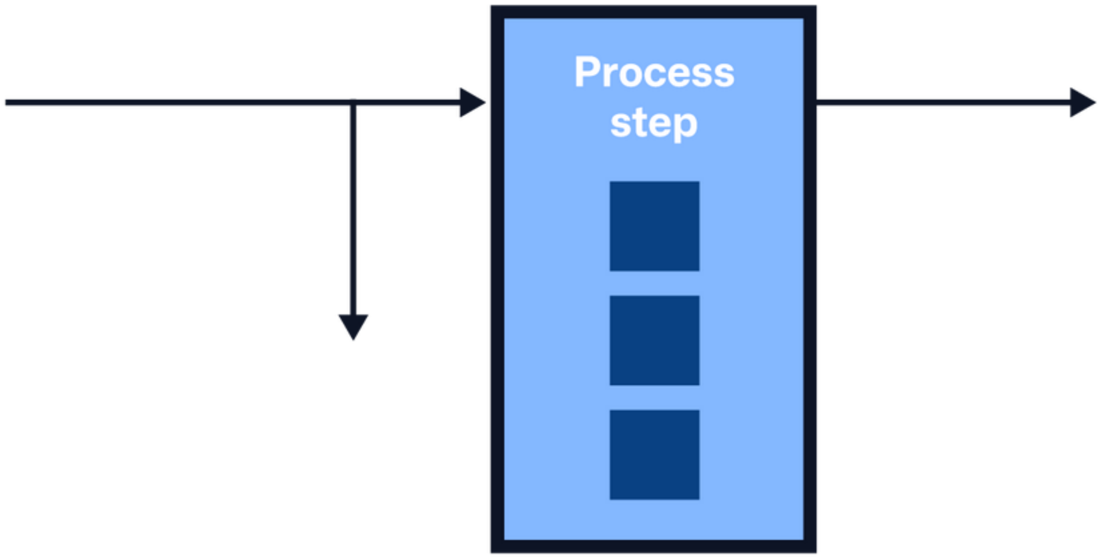
Implications:

- Fraction of customers NOT getting the service: P_m
- Fraction of customers getting the service: $1 - P_m$

Throughput losses: A calculator

Number of servers	3
Average interarrival time	3
Average processing time	2

Average utilisation	21.66%
Loss probability	2.55%



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Q8.8

For the next two questions, consider the following scenario. During the period of final exams, the number of requests for escort services increases to 19.2 per hour (one request every 3.125 minutes). The coefficient of variation of the time between successive requests equals one. However, if a student requesting an escort finds out from the dispatcher that their request would have to be put in the queue (ie: All security officers are busy walking other students), the student cancels the request and proceeds to walk on alone.

(c) How many students per hour who called to request an escort end up cancelling their request and go walking on their own?

(d) University security regulations require that at least 80% of the students' calls to request walking escorts have to be satisfied. What is the minimum number of security officers that are needed in order to comply with this regulation?

Q8.8 solution

(c) m = 8

p = 25 mins

a = 3.125 mins

Use erlang Loss calculator: all servers busy with prob: 23.6%

This means that 23.6% * (60/3.125) calls = 4.52 requests are cancelled per hour.

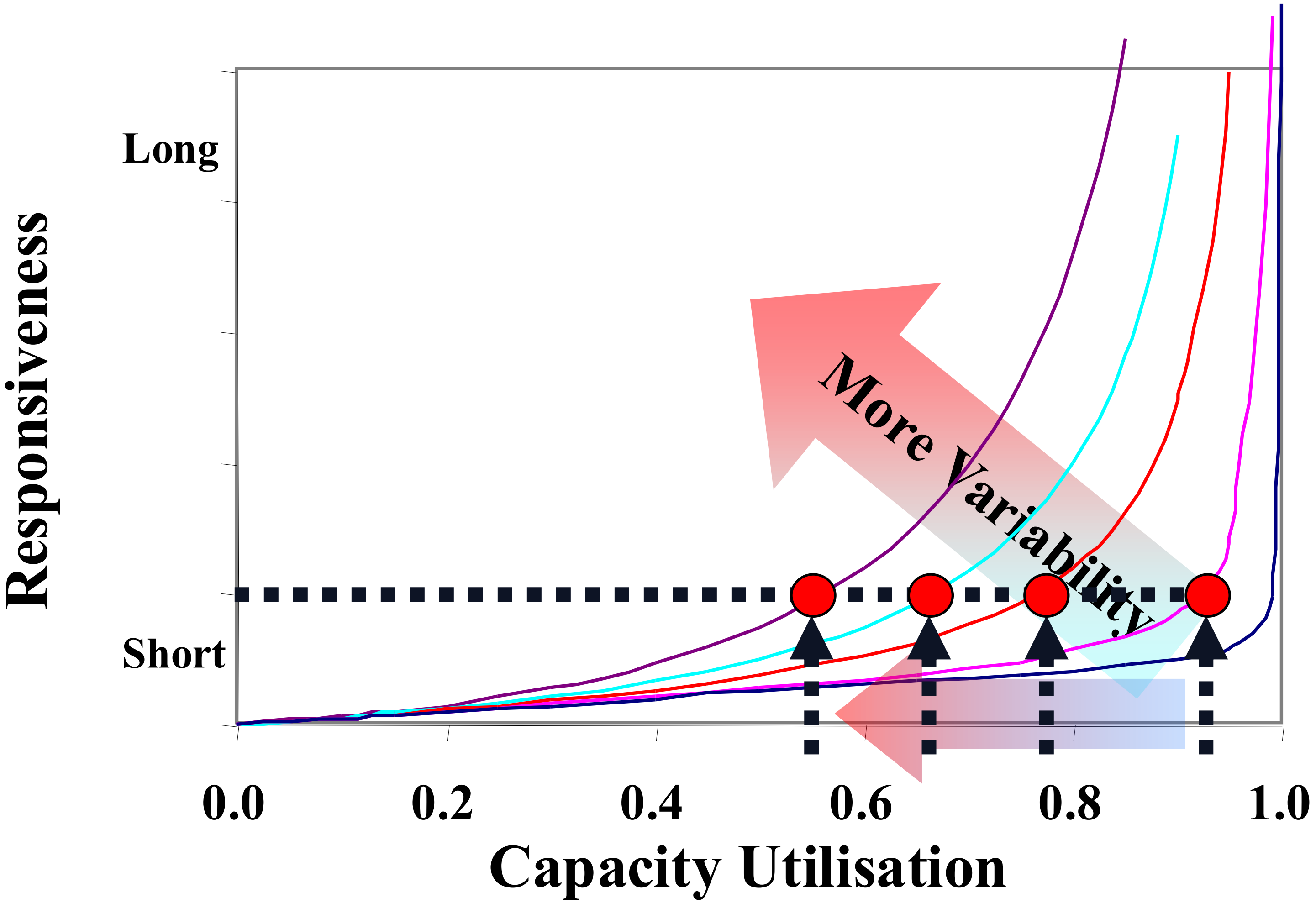
Number of servers	8
Average interarrival time	3.125
Average processing time	25

Average utilisation	76.44%
Loss probability	23.56%

(d) If we try one more officer (i.e. m=9), the loss probability becomes less than 20% as desired.

The university needs at least 9 security officers to meet regulation.

Impact of variability



Other factor that affect throughput loss and loss of goodwill

Throughput and goodwill losses don’t only depend on the length of the waiting times.

It also depends on the customers’ perception of time.

Queue Structure

What type of queue do you prefer?

Queue Structure

Multiple queue structure: Each server has its separate queue.

Single queue structure: all servers attend customers from the same queue.

Queue Structure

There is evidence that single queue structure is generally preferred.

- Provides perception of a faster queue progress.
- More fair.

Perception of Social Justice

First come first serve or priority queues?

How premium customers affect perception of waiting times?

Perception of Social Justice

Many studies have shown how the perception of social justice positively influence customers.

Managerial implications

- Implementation of First Come First Serve;
- Keep premium facilities (or queues) out of sight;

Employee visibility

Studies have shown how customers are positively influenced if the visible employees are serving customers.

Managerial implications:

Visible employees should be serving customers.

Key takeaways

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Many processes experience variability.

Main two types of variability:

- Variability of Demand (way to measure: coefficient of variation of the arrival demand, CV_a)
- Variability of the processing times (way to measure: coefficient of variation of the processing time CV_p)

Waiting times exists even when process utilisation is less than 100%

Basic model allows us to calculate the avg waiting time based on: a, p, CV_a, CV_p and the number of servers.

Managerial insight: to manage a process with high variability, additional capacity is required.

Key takeaways

- Variability can also result in loss of throughput.
- Erlang loss probability: probability that all servers are busy.
- Other managerial actions to combat throughput losses and waiting times:
  - Reduce variability of demand
  - Reduce variability of processing times
  - Merge systems: pooling

Pooling example

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Before pooling

Number of servers	1
Average interarrival time	2.00
Average processing time	1.00
Coeff. of Variation for arrival times	1.00
Coeff. of Variation for processing time	1.00

Average server utilisation	**50.00%**
Waiting time	**1.00**

After pooling

Number of servers	2
Average interarrival time	1.00
Average processing time	1.00
Coeff. of Variation for arrival times	1.00
Coeff. of Variation for processing time	1.00

Average server utilisation	**50.00%**
Waiting time	**0.37**

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